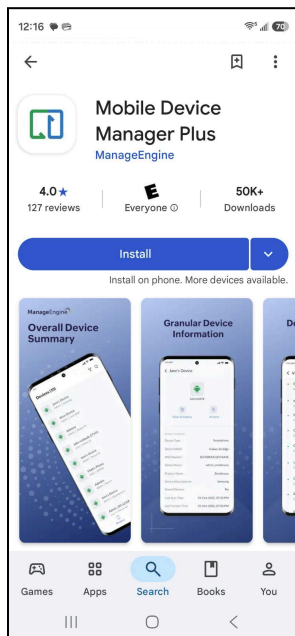
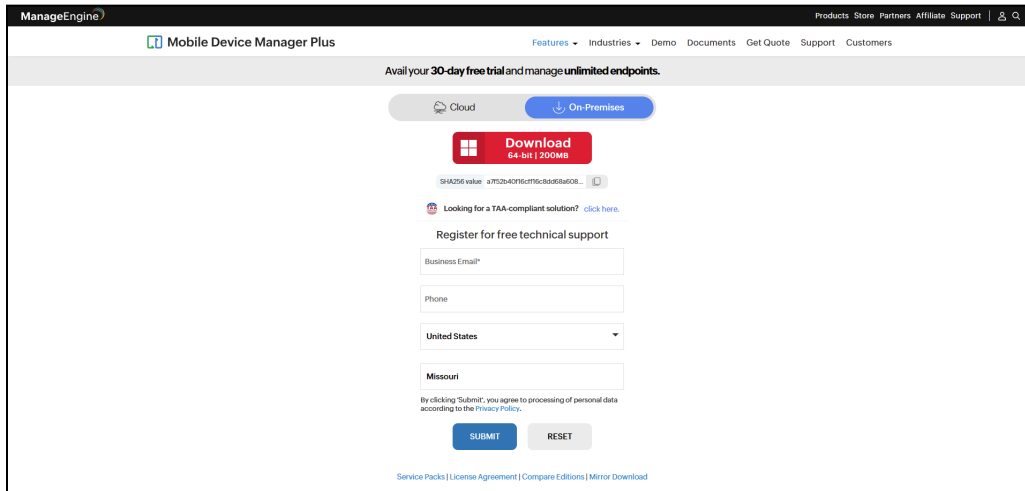


Mobile Device Management Homelab Documentation

Objective 1: Install Mobile Device Manager Plus on the host device and client devices. The MDM was installed on my workstation as the host and it was installed on my mobile devices - my current phone and my old phone - to be clients.



Objective 2: Enroll mobile devices in the MDM. A QR code was generated and to be scanned by the mobile devices. The mobile devices scanned this QR code and were successfully enrolled in the MDM. NAT settings (censored) were enabled to allow communication between devices.

ManageEngine Mobile Device Manager Plus

Home Device Mgmt Inventory Enrollment Reports Admin Support

Enroll Devices Users Self Enrollment Enrollment Settings Directory Services

Apple Apple Enrollment (ABM/ASM) Apple Configurator APNs certificate ME MDM App

Android QR Code Enrollment Zero-touch Enrollment Knox Mobile Enrollment NFC Enrollment Managed Google Play ME MDM App

Windows Enroll Laptop/Surface Pro Azure Enrollment (AutoPilot)

Staged Devices QR Code Template

QR Manual Template Create New Template

9.0 and later 6.0 to 9.0

1. Activate & Enroll
On a new/factory reset device, Tap the Welcome screen 6 times to access the device camera and scan the QR code shown here.

Optionally Configure Wi-Fi to skip this step while setting up the device

2. Provision
You can manually assign users from Staged tab to complete enrollment
Management Type : Full Device Management

View list of devices enrolled through QR Code Enrollment.

Copy QR Code Data URL

ManageEngine Mobile Device Manager Plus

Home Device Mgmt Inventory Enrollment Reports Admin Support

Setting up MDMP NAT Settings Proxy Settings Mail Server Failover Server

Global Settings User Administration Server Settings Rebranding Server Migration Review MDMP Need more services to migrat...

Privacy Settings Device Privacy Server Privacy Terms of Use

Security Settings Security Settings Manage SSL Certificate Secure Gateway Server Export Settings

Configure firewall

To ensure a smooth connection between the Endpoints and the Central Server, open Inbound ports 9383 and Outbound Port 443 on the Central Server. Learn More

To ensure continued management of devices, use an FQDN as changes in the IP address are automatically mapped to the FQDN. Learn more

NAT Settings

Central Server (Private IP Address and Ports)

Public FQDN

Server Port 9383

NAT Device (Public FQDN and Ports)

Public FQDN

Server Port 9383

Secure communication to Central Server

Use Secure Gateway Server to secure all inbound communications to Central server.

Watch Demo Configure Now

Note : All Communications are in HTTPS mode. You can also use third party certificates to ensure data security. Learn More: How to use third party certificates

Save

How it works?

Apple Push Notification Service (iOS) / Google Cloud Messaging Service (Android) / Windows Notification Service (Windows)

DMZ

Wake Up Device

1

2

Objective 3: Add users and assign them to devices and groups. The admin user was assigned to my currently in-use phone and a user using an old non-business email of mine was assigned to my old phone. One group was created for each device just to familiarize myself with group concepts, even though I don't actually have enough devices for groups. My old phone is my dirty "cooking phone" for using in the kitchen so I named the groups "kitchen" and "not kitchen".

The screenshot shows the 'Users' page in the ManageEngine Mobile Device Manager Plus interface. The left sidebar contains navigation options: Enroll, Apple, Android, and Windows. The main content area displays a table of users with columns: Username, Display Name, Email, Domain Name, Associated Devices, and Action. Two users are listed: 'admin' and 'redgroupclan'. The 'admin' user is associated with 1 device, and the 'redgroupclan' user is also associated with 1 device. The page includes a 'Rows per page: 25' dropdown and a '1 - 2 of' pagination indicator.

Username	Display Name	Email	Domain Name	Associated Devices	Action
admin	admin	--	Local User	1	[icon]
redgroupclan	redgroupclan	redgroupclan@gmail.com	Local User	1	[icon]

The screenshot shows the 'Groups' page in the ManageEngine Mobile Device Manager Plus interface. The left sidebar contains navigation options: Manage, Apps, Tools, Conditional Access, and Geofencing. The main content area displays a table of groups with columns: Group Name, Group Type, Members, Profile Count, Apps Count, Content Count, Modified by, Modified Time, Action, and Group Category. Two groups are listed: 'Kitchen' and 'Not kitchen'. Both groups are 'Device Group' type, have 1 member, and were modified by 'admin' on Sep 14, 2026. The page includes a 'Rows per page: 25' dropdown and a '1 - 2 of' pagination indicator. Below the table, there is a 'Quick Links' section with a list of links and a 'More | Roadmap' button.

Group Name	Group Type	Members	Profile Count	Apps Count	Content Count	Modified by	Modified Time	Action	Group Category
Kitchen	Device Group	1	0	0	0	admin	Sep 14, 2026 11:01 PM	[icon]	Static
Not kitchen	Device Group	1	0	0	0	admin	Sep 14, 2026 10:40 PM	[icon]	Static

Quick Links

How To's

1. How to associate profiles to groups?
2. How to associate profiles to devices?
3. How to distribute apps to groups?
4. How to remove/disassociate profiles from devices?
5. How to uninstall/remove app from group?
6. How to uninstall/remove app from devices?

[More](#) | [Roadmap](#)

Objective 4: Verify the devices are successfully enrolled by distributing an announcement that appears on all enrolled devices.

ManageEngine
Mobile Device Manager Plus

Your Trial expires in 17 days. [License](#) | [Get Quote](#) Build Version : 11.2.2605.31

Home Device Mgmt Inventory Enrollment Reports Admin Support

Manage
Groups & Devices
Profiles
Telecom Expense Mgmt
Certificates
Alerts
Content Management
Automate OS Updates

Apps
App Repository
App Update Policy

Tools
Announcements
Remote Control

Conditional Access
Exchange
Office 365
Office 365 MAM policy

Geofencing
Fence Policy
Fence Repository

Modify An Announcement

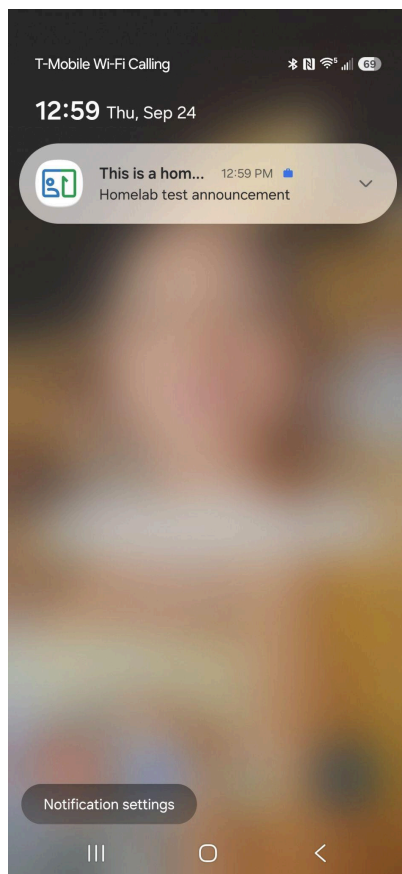
Announcement Title*

Message*

Select

B *I* U **≡** **≡** **A**

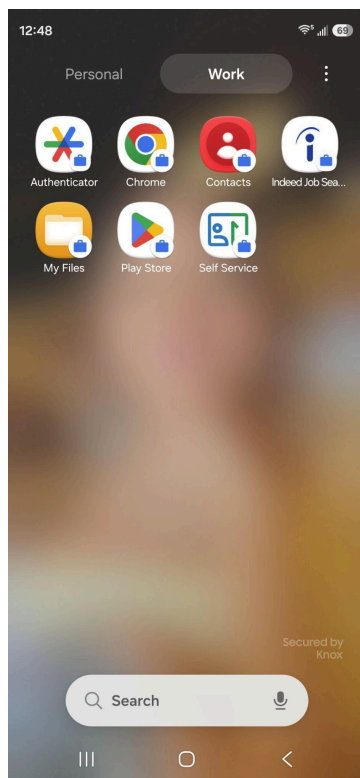
Homelab test announcement



Objective 5: Force install apps onto the managed phones to familiarize myself with installing apps from the app repository. Using the MDM controls, I successfully installed a couple apps from the Play Store and set them to be uninstallable by the user. The apps chosen were Indeed Job Search, just as a test, and Google Authenticator, to provide OTPs for a future security policy I would be implementing.

The screenshot shows the ManageEngine Mobile Device Manager Plus web interface. The top navigation bar includes links for Home, Device Mgmt, Inventory, Enrollment, Reports, Admin, and Support. A sidebar on the left lists various management categories like Manage, Groups & Devices, Profiles, and Apps. The main content area is titled 'Install ME MDM app silently in iOS devices' and features a table of installed apps. The table has columns for App Name, Platform Type, App Type, Latest Version, Supported Devices, Modified Time, and Action. Two apps are listed: 'Indeed Job Search' and 'Google Authenticator'. Below the table, there are 'Quick Links' and a 'More | Roadmap' link.

App Name	Platform Type	App Type	Latest Version	Supported Devices	Modified Time	Action
Indeed Job Search	Android	Android Store App	295.0.0	Smartphone, Tablet	Sep 17, 2026 06:21 PM	
Google Authenticator	Android	Android Store App	7.2	Smartphone, Tablet	Sep 13, 2026 10:13 PM	



Objective 6: Create an app update policy to control when apps would be updated. I created a new policy specifically for the Indeed Job Search app. Updates were set to be manually approved during distribution during a scheduled window of 5AM-9AM daily.

Update Policy

×

Policy Name *

App Update Policy

App Update Policy For

☐ All Apps

☐ All Store Apps

☒ Specific Apps

Indeed Job Search

Select the apps to be included in the policy

1

Approve Updates from Store ?

☐ Automatically When Detected

☒ Manually during distribution

Distribute Approved Updates

☐ Immediately

☒ Scheduled Window

Scheduled Window

App Update Frequency

☒ Daily

☐ Weekly

Preferred Deployment Time *

Start at

05:00

End at

09:00

(GMT-5:00) Central Daylight Time (America/... ▾

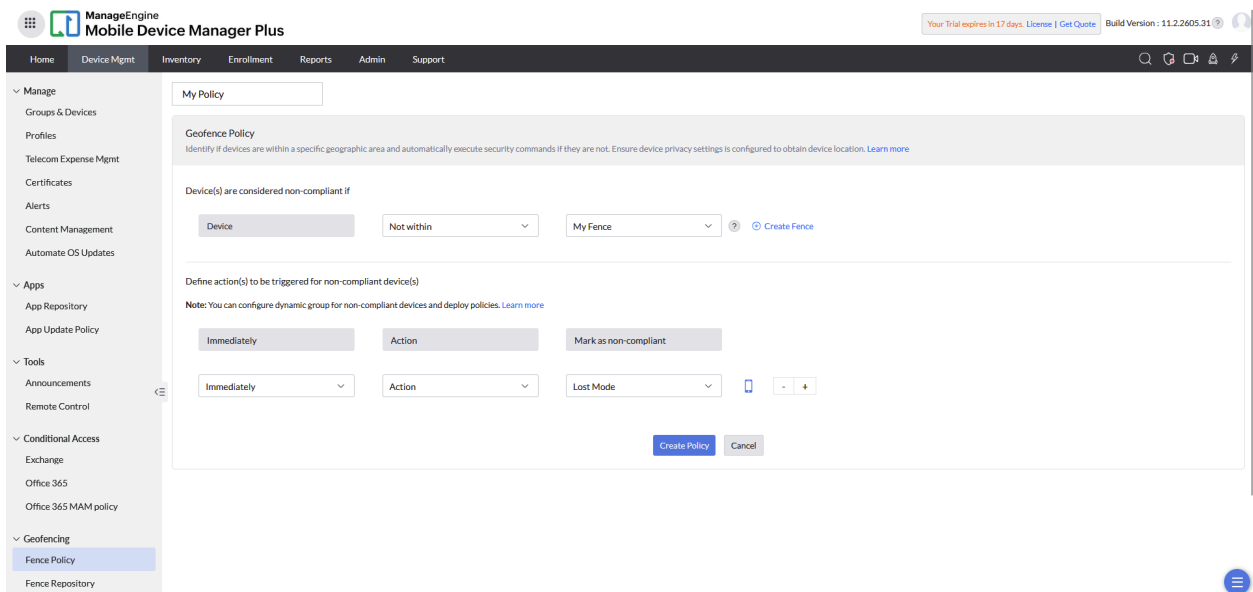
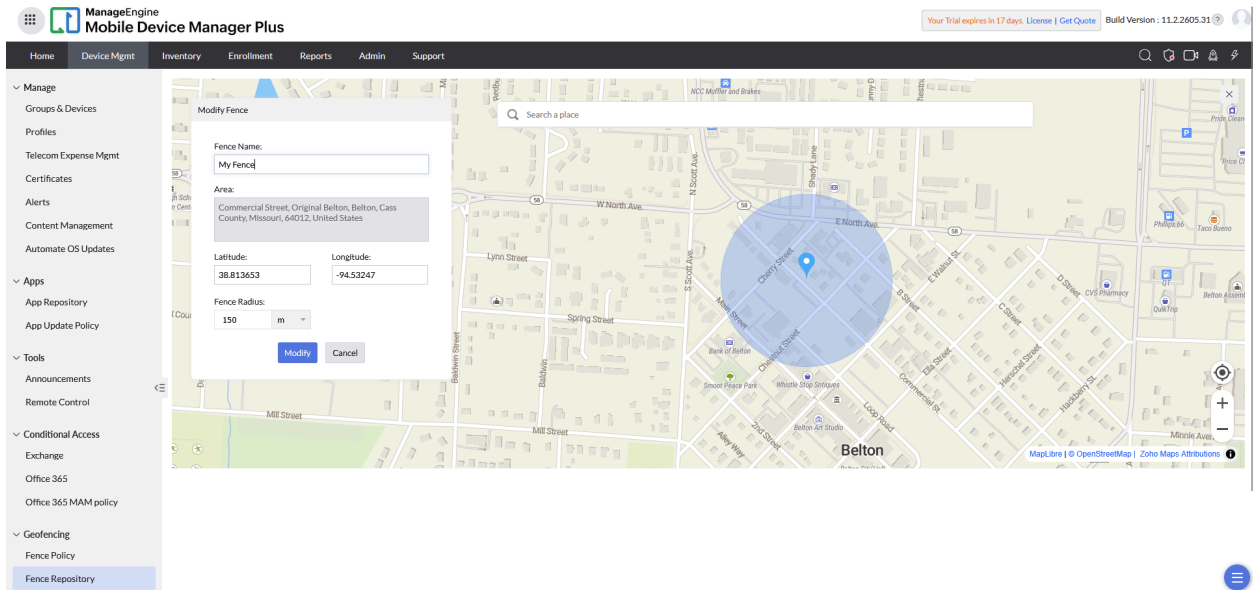
ⓘ

The ongoing and future updates will follow the modified configurations. Updates that are distributed already cannot be changed. The changes made to this policy, have resulted in conflicts for associated devices. [Learn More](#)

Modify

Cancel

Objective 7: Familiarize myself with geofencing controls. I set up a fence for an area around where I live. After that, I created a fence policy that would immediately put the device in Lost Mode (locking it server-side) if the device were to leave the fence. The user would get a notification if they were leaving the fence.



Objective 8: Create a policy to automate OS updates. I know that OS updates can be a tricky thing to manage because an update might break something. For this reason, I delayed them for 3 days and set the deployment schedule to the last Saturday of the month.

Policy details

×

Applicable for devices running Android 6.0 or later versions and provisioned as [Device Owner](#).

Policy Name *

Android Policy

Deploy OS on devices *

☐ Immediately ☒ Delay for

3

 Day(s)

Deployment Schedule

Preferred Weeks for Deployment *

First Second Third Fourth **Last**

Preferred Days for Deployment *

Sun Mon Tue Wed Thu Fri **Sat**

Deployment Window *

06:00

 to

09:59

 [hh:mm][24-hour format]

Note: Deployment happens within the time limit specified. If it cannot be completed, it gets resumed during the subsequent deployment window.

Update Freeze Periods

Start Date

1

Jan

End Date

15

Jan

−

+

Note: You can configure a freeze period to suspend OS updates for up to 90 days. Two freeze periods cannot overlap each other and must be separated by a minimum of 60 days.

☒ Notify users regarding updates

Message Title

OS update pending!

Message Contents

Your organization has configured an OS update for this device.

Objective 9: Create a security profile. For this, it is hard to show every single detail of the policy as there are hundreds of different options to configure. To summarize, I went with a profile that errs on the side of what I would imagine a strict corporate security policy would look like.

The screenshot displays the 'Configure Profile' interface for an Android profile in the ManageEngine Mobile Device Manager Plus. The interface is divided into three main sections: a left sidebar for navigation, a central configuration area, and a right sidebar for a live preview.

Left Sidebar (Navigation):

- Passcode
- Restrictions
- Workspace Security
- Wi-Fi
- VPN
- Email
- Exchange ActiveSync
- EFPP
- Kiosk
- Wallpaper
- Asset Tag Information
- Sound Settings
- Global HTTP Proxy
- Certificate
- SCEP
- Web Shortcuts
- Web Content Filter

Central Configuration Area:

- Device Functionality:**
 - Security: Camera (Allow/Restrict), Video Recording (Allow/Restrict), Microphone (Allow/Restrict), Audio Recording (Allow/Restrict), Firmware Recovery (Allow/Restrict), OS upgrade (Allow/Restrict), Screen Capture (Allow/Restrict), Smart Clip Mode (Allow/Restrict), S Voice (Allow/Restrict), Enforce Storage encryption (Yes/No), Enforce SD Card encryption (Yes/No).
 - Sync and Storage
 - Applications
 - Browser Restriction
 - Network and Roaming
 - Device Connectivity
 - Tethering
 - Location settings
 - Phone
 - Date/Time Settings
 - Display Settings
 - Miscellaneous

Right Sidebar (Live Preview):

- Live Preview:** Shows a preview of the device screen with the following settings:
 - Asset tag text size: Small
 - Background Type: Background Color (selected), Wallpaper
 - Asset Tag Screen Orientation: Lock Portrait (selected), Lock Landscape, Auto Rotate
 - Upload Wallpaper: linkedin.png (selected), Upload a separate image for landscape view
 - Text Color: A (selected), #F0F0F0
 - Select the Properties to be displayed: Username, Device Serial Number
 - Custom message: Asset tag information

Objective 10: Associate the created profiles with different groups. I successfully associated MyProfile with the “Kitchen” group, and another profile with the “Not kitchen” group. Both of my phones now had a profile with security policies being enforced.

ManageEngine Mobile Device Manager Plus

Home Device Mgmt Inventory Enrollment Reports Admin Support

Manage

- Groups & Devices
- Profiles
- Telecom Expense Mgmt
- Certificates
- Alerts
- Content Management
- Automate OS Updates

Apps

- App Repository
- App Update Policy

Tools

- Announcements
- Remote Control

Conditional Access

- Exchange
- Office 365
- Office 365 MAM policy

Geofencing

- Fence Policy
- Fence Repository

Groups

+ Create Group X Delete Group Action 1 Selected

Group Name	Group Type	Configuration Profile	File Count	Apps Count	Content Count	Modified by	Modified Time	Action	Group Category
☒ Kitchen	Device Group	Telecom Expense	0	0	0	admin	Sep 14, 2026 11:01 PM		Static
☐ Not kitchen	Device Group		0	0	0	admin	Sep 14, 2026 10:40 PM		Static

Rows per page: 25

Quick Links

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6. How to uninstall/remove app from devices?

More | Roadmap

ManageEngine Mobile Device Manager Plus

Home Device Mgmt Inventory Enrollment Reports Admin Support

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- Exchange
- Office 365
- Office 365 MAM policy

Geofencing

- Fence Policy
- Fence Repository

Profiles > MyProfile

MyProfile Published

Created By : admin | Last Modified By : admin

Profile Details Associated Groups (1) Associated Devices (1)

Action

Group Name	Members	Version	Associated Time	Associated by	Status	Remarks	Action	Assigned Action
☐ Kitchen	1	1	Sep 24, 2024 01:25 PM	admin	Ready To Execute	--		Profile Association

Rows per page: 25

Objective 11: Explore other useful features of the MDM. This included geotracking, scanning devices for compliance, battery and network tracking, reports, remote lock & wipe, and the list goes on. I dare not waste the readers time by documenting every single feature out of the HUNDREDS I configured.



Set Up Geo-Tracking

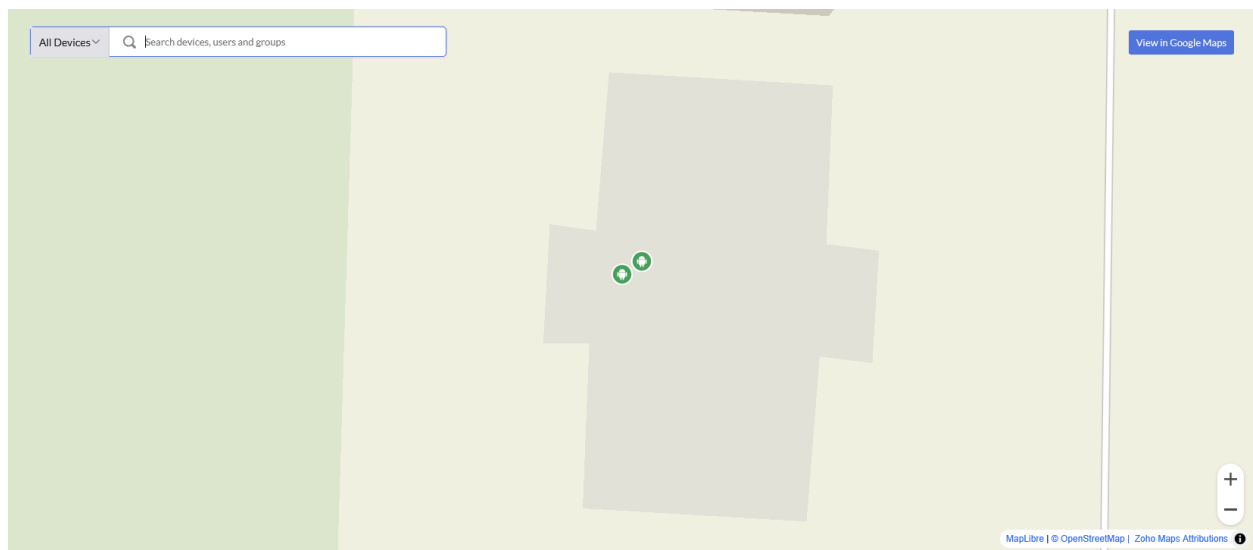
Setting up Geo-Tracking allows you to monitor and secure your devices by tracking their locations efficiently.[Learn more](#)

Track devices * ☒ Always ☐ When lost [?](#)
[Learn how to prevent users from disabling location services.](#)

Apply Geo-Tracking for ☒ All devices ☐ Specific group(s) ☐ All except selected group(s)

Location History [?](#) ☐ Enable ☒ Disable

[Save Changes](#) [Disable](#)



[Home](#) [Device Mgmt](#) [Inventory](#) [Enrollment](#) [Reports](#) [Admin](#) [Support](#)

[Predefined Reports](#) [Schedule Reports](#) [Query Reports](#)

[Need more enhancements for achieving compliance?](#)

Create Scheduled Reports

Scheduler Name *

Description

Select Reports

Reports

- ☒ Reports
 - ☒ App Installation summary - App installation count
 - ☒ App Installation summary - List of apps by devices
 - ☒ Devices with/without Specific App
 - ☒ Blocklisted Apps Summary [?](#)
- ☐ Network Performance
 - ☒ Devices with blocklisted apps
- ☒ Data Usage by Devices
- ☒ Data Usage by Groups
- ☒ New App detected
- ☒ Devices by Model

Selected Reports (26)

- Data Usage by Devices
- Data Usage by Groups
- New App detected
- Devices by Model
- Device SIM details
- Device Battery History
- Devices by Enrollment Time
- Inactive Devices
- Recent Locations
- Geo-tracking status
- Rooted Devices

Specify Delivery Format and Recipients

Export settings [Export Setting to secure personal data is not configured. \[Configure Now\]\(#\)](#)

Report Formats ☒ PDF ☐ XLSX ☐ CSV

Receive emails even when no data is available ☒ Yes ☐ No

Specify Delivery Format and Recipients

Conclusion: While learning the ins-and-outs of Mobile Device Management, I learned just how robust a MDM can be. The configuration options are practically endless. I found the experience not only interesting, but also empowering. It was fun configuring options and seeing those configurations actually get pushed to my phones. I learned a great deal about mobile device management in this homelab. It made me realize how much I would like a MDM full-time. The geotracking and forced phone alarm were especially useful for finding my phone when I could not find it. All in all, I spent dozens of hours configuring this lab and I was engaged the whole time.

My greatest challenge with this homelab was that some of the features required being connected to enterprise-level resources that I did not have access to and were beyond the scope of this homelab. I look forward to overcoming that challenge in an authentic enterprise environment in the future!